PENETRATION TESTING PROJECT PLAN

Custom Vulnerable Linux Machine
& Penetration Testing Lab

Four-Week Build, Exploitation & Reporting Plan

Project Duration: October 8 – November 6
Team Size: 5 Members

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1. Executive Summary

Our team builds a vulnerable Linux virtual machine from scratch using the topics we studied in class. The lab covers four main steps: scanning the network, getting an initial shell, escalating privileges to root, and keeping access with persistence. After finishing the build, we hack the machine from Kali Linux, take screenshots of every step, and write a clear report on how to fix each security issue.

1.1 Scope & Rules of Engagement

All testing runs only inside our private virtual network (Host-Only / NAT). We do not touch any external or real-world networks. All vulnerable files, users, and passwords are created by our team for practice only.

2. Technologies & Tools

Category	Tools
System & Virtualization	Ubuntu Server, VMware / VirtualBox
Network & Scanning	Nmap, Wireshark
Exploitation	Metasploit Framework (msfconsole), Meterpreter, msfvenom
Privilege Escalation	Linux permissions, Sudo misconfiguration, Cron jobs, local_exploit_suggester
Documentation	GitHub, Markdown, MS Word

3. Timeline & Weekly Tasks (4-Week Plan)

Week 1 — Lab Setup & Target Configuration *(Oct 8 – Oct 15)*

- Install Ubuntu Server VM on an isolated network.
- Create system users (a normal user and root).
- Configure basic network settings and test connectivity.

Deliverable: *Working base machine and network setup document.*

Week 2 — Initial Access & Service Exploitation *(Oct 16 – Oct 22)*

- Install target services (e.g. a vulnerable FTP or web service).
- Create payloads using msfvenom and test listeners in msfconsole.
- Capture network traffic during scanning using Wireshark.

Deliverable: *Successful initial shell (Meterpreter) and Wireshark capture notes.*

Week 3 — Privilege Escalation & Persistence *(Oct 23 – Oct 29)*

- Add a vulnerable cron job with weak file permissions.
- Misconfigure /etc/sudoers to allow root access for a specific command.
- Test escalating to root using local_exploit_suggester and manual Linux commands.

- Set up a persistence backdoor and place user.txt and root.txt flags.

Deliverable: *Working root escalation path and verified flags.*

Week 4 — Final Testing, Report & OVA Export *(Oct 30 – Nov 12)*

- Run a full penetration test from Kali Linux, screenshotting every step.
- Fix the misconfigurations and write remediation steps.
- Export the final machine as an .ova file and update the GitHub repository.
- Write the final penetration testing report and prepare the presentation.

Deliverable: *Complete pentest report, walkthrough guide, and exported .ova file.*

4. Team Roles (5 Members)

Member	Primary Role	Assigned Tasks
Mohamed Ahmed Tawfiq Abdelkader (Team Lead)	VM & Network Setup	Installing Ubuntu Server, network isolation, and managing the GitHub repository.
Ahmed AbdElraouf Mohamed	Scanning & Analysis	Running Nmap scans, testing evasion options, and capturing packets with Wireshark.
Youssef Ali Mohammed Alamrousy	Foothold & Exploitation	Setting up the vulnerable entry service, creating payloads with msfvenom, and handling listeners.
Walid Khalaf ahmed	Privilege Escalation	Configuring vulnerable cron jobs, editing sudoers, and setting up persistence and flags.
Marwan Mahmoud Ibrahim	Reporting & Remediation	Taking screenshots, writing the step-by-step walkthrough, and drafting the final security report.